

Ahmed Hassan Eldemery

AI Engineer — Machine Learning Engineer

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Professional Summary

AI Engineer and Machine Learning professional building AI agents, computer vision systems, and intelligent automation solutions. Skilled in backend API development, computer vision, and prompt engineering using Python, FastAPI, Scikit-learn, OpenCV, and MongoDB. Passionate about transforming real-world problems into scalable AI products, with hands-on experience delivering production-grade systems spanning multimodal AI classification, SEO automation, and business process automation.

Professional Experience

Artificial Intelligence Engineer

Jul 2026 – Aug 2026

Howeyah Digital Marketing

Remote

- Improved Howeya SEO-AI workflows for WordPress analysis, recommendations, media metadata, and Alt Text automation.
- Built multi-stage validation covering live content, SEO rules, freshness checks, and traceable evidence.
- Strengthened crawling and data processing reliability, with 21/21 PostgreSQL tests and 394/394 regression tests passing.

Artificial Intelligence Specialist

Apr 2026 – Jul 2026

Try GC (formerly Grand Community)

Mansoura, Egypt – On-site

- Built a multimodal influencer classification system processing **60,000+ profiles**, combining computer vision (face/video analysis) and speech models (TalkNet, Whisper) to auto-classify creators.
- Developed a Manifest V3 Chrome Extension and FastAPI backend to record meetings, transcribe Arabic conversations via Whisper, and extract structured marketing insights using Llama-3.3 via Groq.
- Designed LLM integration pipelines and prompt workflows, reducing manual work across the company's core database.
- First member of the AI team; helped recruit and onboard university colleagues onto the team.

Machine Learning Engineer Trainee

Jun 2025 – Dec 2025

Huawei & Mansoura University Collaboration

Mansoura, Egypt

- Engineered end-to-end ML pipelines (Python, Scikit-learn), improving model accuracy by ~12% via feature engineering.
- Trained and evaluated supervised and unsupervised models on real-world datasets.

Machine Learning Engineer Intern

Jul 2025 – Aug 2025

National Telecommunication Institute (NTI)

Mansoura, Egypt

- Developed a predictive analytics model using Scikit-learn to forecast e-commerce customer spending behavior.
- Executed end-to-end ML data pipelines, enabling data-driven insights from historical sales data.

Technical Skills

- **AI & LLMs:** LLM Integration, Prompt Engineering, LangChain, RAG, Groq API, Llama 3, OpenAI API
- **Computer Vision & Speech:** OpenCV, YOLOv8, PaddleOCR, MediaPipe, TalkNet, Whisper, CLIP
- **Machine Learning & Data:** Scikit-learn, XGBoost, Pandas, NumPy, Regression, Classification, Clustering, Feature Engineering
- **Backend & Systems:** FastAPI, Flask, React, REST APIs, WebSockets, JWT, PostgreSQL, MongoDB, PGVector, Qdrant
- **Tools & Architecture:** Python, SQL, Git, Docker Compose, Microservices, Chrome Extensions (MV3), WordPress REST API

Education

Bachelor of Computer Science (Minor in Artificial Intelligence)

Oct 2023 – Jul 2027 (Expected)

Mansoura University

Mansoura, Egypt

- **Relevant Coursework:** Machine Learning, Computer Vision, Data Structures, Linear Algebra, Probability & Statistics

Key Projects

Howeya SEO-AI (Howeyah)

Python, FastAPI, PostgreSQL, WordPress REST API, Pytest, Docker

- Improved WordPress SEO analysis, media/Alt Text automation, crawling, and evidence-backed recommendation workflows.
- Built multi-stage validation and strengthened reliability, with 21/21 PostgreSQL tests and 394/394 regression tests passing.

Influencer Video Classifier (Try GC)

Python, OpenCV, TalkNet, Whisper

- Engineered a multimodal pipeline combining computer vision and speech models to classify creators by content type and speaking style.
- Automated classification workflows across a database of **60,000+ creator profiles**.

AI Meeting Assistant (Try GC)

FastAPI, Chrome Extension (MV3), Groq API, Whisper

- Engineered a Manifest V3 Chrome Extension utilizing offscreen documents to bypass service worker constraints for real-time audio mixing and secure microphone capture.
- Built a stateless FastAPI backend with strict Pydantic validation to process audio chunks via Groq's Whisper-large-v3 and extract structured Arabic marketing insights using Llama-3.3 in JSON mode.

VoltIQ – Smart Electricity Platform

FastAPI, Flask, YOLOv8, PaddleOCR, XGBoost, PGVector, Docker

- Architected a microservices AI platform combining computer vision OCR, predictive ML, and a RAG document assistant.
- Built an OpenCV/PaddleOCR pipeline to read LCD 7-segment displays and trained XGBoost models for anomaly detection and balance depletion forecasting.
- Deployed a FastAPI RAG microservice utilizing PGVector/Qdrant and LLMs (Gemini/OpenAI) for context-grounded Q&A.

YAQIZ – AI Workplace Safety & Fatigue Monitoring

FastAPI, React, YOLOv8, MediaPipe, Docker

- Built a full-stack platform combining YOLOv8 PPE detection with ByteTrack worker tracking and MediaPipe fatigue analysis.
- Engineered WebSocket pipelines for live video streaming and containerized the full stack using Docker for scalable deployment.

Digital Employee – AI MSME Sales Assistant

Python, Flask, OpenAI API, MongoDB, JWT

- Built a bilingual LLM chat assistant converting conversations into structured orders with auto-generated IDs.
- Implemented role-based access, low-stock alerts, and config-driven workflows for 10+ business sectors.

ErgoAI – Intelligent Health & Productivity Assistant

Python, OpenCV, Computer Vision

- Developed an AI-powered desktop app utilizing OpenCV to detect blink rate, yawning, and posture at ~20 FPS.
- Achieved **3rd Place** in the Mansoura University Computer Vision Course and **3rd Place** in the university-level Ibtikar 8 Innovation Competition.

Hackathons & Competitions

- **Huawei Developer Competition (Northern Africa):** Advanced to the Top 50 teams with Team Sanad (2025).
- **First AI Summit Week Hackathon (Mansoura University):** Exhibited the Digital Employee prototype alongside teams from 16 universities (2026).
- **Egypt Hackathon 4.0 (GDG Delta):** Built the initial YAQIZ prototype among 120+ teams and 600+ developers (2025).

Honors & Certificates

- **Awards:** 3rd Place – Ibtikar 8 Innovation Competition (2025) — 3rd Place – Mansoura Computer Vision Course (Apr 2025) — Digitopia Semi-Finalist (Jun 2025).
- **ALX Africa Data Science Program:** 15-month intensive track in Data Analytics and Machine Learning (June 2026).
- **Certifications:** NVIDIA (Building LLM Applications, AI for All) — Intro to Machine Learning (Grade: Excellent) — Applied Data Science & Python.
- **Digital Egypt Pioneers Program (DEPI):** AI & Data Science - Microsoft Machine Learning Engineer, sponsored by MCIT (Nov 2025 - Jul 2026).